

PELLSTON, MI, USA

WMO: 727347

Lat: 45.564N

Lon: 84.793W

Elev: 215

StdP: 98.77

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14841

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.6	-20.7	-27.5	0.3	-24.3	-23.5	0.5	-20.1	11.9	-6.1	10.9	-5.7	0.8	310	0.608

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.2	30.4	21.5	28.8	20.6	27.4	19.7	23.1	28.3	22.0	26.6	21.1	25.3	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.5	25.6	20.5	15.5	24.7	19.5	14.6	23.6	69.5	28.6	65.6	26.5	62.0	25.4	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.0	8.2	-30.3	33.6	4.0	1.4	-33.1	34.6	-35.5	35.4	-37.7	36.2	-40.6	37.2		
(5)			-30.5	24.9	3.8	1.0	-33.2	25.7	-35.5	26.3	-37.6	26.9	-40.4	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.2	-7.7	-7.5	-2.6	4.5	11.5	16.9	19.2	18.4	14.5	8.1	2.1	-3.6		
	DBStd	10.85	6.44	6.52	6.57	4.98	4.70	4.13	3.35	3.26	4.42	4.80	4.83	5.45		
	HDD10.0	2413	550	489	393	176	36	2	0	0	9	96	242	421		
	HDD18.3	4574	808	723	649	416	218	75	30	40	129	318	488	679		
	CDD10.0	1038	0	0	3	11	83	209	286	260	145	38	3	0		
	CDD18.3	157	0	0	0	1	7	33	57	42	16	2	0	0		
	CDH23.3	1857	0	0	4	11	132	421	658	438	173	20	0	0		
	CDH26.7	460	0	0	1	2	32	120	174	92	37	3	0	0		
(14) Wind	WSAvg	3.4	3.9	3.9	3.7	4.0	3.5	3.0	2.9	2.7	2.8	3.4	3.8	3.8		
(15) Precipitation	PrecAvg	815	56	34	55	63	76	66	66	83	99	79	79	61		
	PrecMax	1083	108	90	196	147	178	174	155	162	231	184	149	128		
	PrecMin	615	11	7	8	25	10	14	5	18	17	23	23	28		
	PrecStd	120	27	19	42	25	32	41	39	37	50	40	30	26		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	9.7	19.6	24.5	30.0	32.5	32.6	31.2	30.2	26.1	18.9	12.2		
		MCWB	5.5	7.3	13.5	15.0	19.8	22.9	22.9	22.2	21.4	18.4	13.1	10.6		
	2%	DB	4.5	6.1	13.2	20.5	27.0	29.8	30.2	29.0	27.4	22.4	14.4	7.4		
		MCWB	2.7	3.3	9.2	12.5	17.8	20.8	21.7	21.3	20.0	16.8	11.0	5.4		
	5%	DB	2.5	3.0	9.5	17.2	24.1	27.7	28.6	27.4	25.2	19.4	11.9	5.2		
		MCWB	1.4	1.3	6.0	9.9	15.8	19.6	20.7	20.1	18.8	15.0	8.7	3.4		
	10%	DB	1.1	1.5	6.4	13.9	21.3	25.7	27.1	25.9	23.1	16.5	9.5	3.4		
		MCWB	0.2	0.2	3.3	7.9	14.3	18.2	19.8	19.2	17.9	13.1	7.0	2.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	7.8	14.3	16.6	21.1	24.0	25.0	23.9	22.7	20.4	14.0	10.6		
		MCDB	7.5	9.1	17.7	22.3	27.7	30.8	30.4	28.4	27.6	23.7	17.3	11.7		
	2%	WB	2.9	3.8	10.0	13.5	19.2	22.3	23.3	22.6	21.2	17.6	11.4	5.7		
		MCDB	4.3	5.8	12.9	19.3	25.2	28.2	28.2	27.0	25.2	21.3	14.0	6.9		
	5%	WB	1.4	1.7	6.2	10.7	17.3	20.8	22.0	21.6	20.1	15.3	9.1	3.7		
		MCDB	2.4	3.0	9.2	15.8	22.0	25.5	26.5	25.5	23.8	19.0	11.4	5.0		
	10%	WB	0.2	0.3	3.6	8.4	15.3	19.5	20.9	20.5	18.7	13.3	7.2	2.2		
		MCDB	1.1	1.4	6.0	13.4	20.2	24.0	25.2	24.2	22.6	16.1	9.3	3.4		
(35) Mean Daily Temperature Range	MDBR		8.6	10.4	11.4	12.3	13.9	13.9	13.2	13.1	13.0	10.6	8.0	7.1		
	5% DB	MCDBR	8.4	10.6	14.5	18.7	17.9	16.2	15.4	14.7	15.4	14.9	12.1	8.5		
		MCWBR	7.4	8.7	10.3	11.6	9.7	8.5	7.9	7.9	8.9	9.7	9.2	7.3		
	5% WB	MCDBR	7.7	9.5	13.8	16.8	14.9	13.9	12.3	11.7	12.9	13.5	11.2	7.8		
		MCWBR	6.9	8.1	10.2	11.2	9.0	7.9	6.9	6.9	8.2	9.4	9.0	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.292	0.315	0.355	0.375	0.391	0.389	0.382	0.369	0.335	0.302	0.278		
	taud		2.310	2.337	2.393	2.340	2.362	2.348	2.376	2.394	2.417	2.496	2.522	2.452		
	Ebn at Noon		798	890	918	903	893	876	873	867	848	833	808	801		
	Edh at Noon		79	92	102	117	119	121	117	110	99	80	64	62		
(44) All-Sky Solar Radiation	RadAvg		1.15	2.08	3.52	4.62	5.66	6.14	6.18	5.25	3.99	2.32	1.23	0.84		
	RadStd		0.13	0.21	0.36	0.48	0.48	0.45	0.23	0.32	0.33	0.27	0.17	0.08		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page